Automated Detection of Intracranial Hemorrhage on Head Computed Tomography with Deep Learning

Jiajie He
Department of Mechanical Engineering,
École Centrale de Lille
Villeneuve-d'Ascq, 59493, France
jiajie.he@master.centralelille.fr

ABSTRACT

Intracranial hemorrhage is a serious health problem worldwide requiring rapid and often intensive medical treatment. However, the diagnosis process of intracranial hemorrhage is complicated and often time consuming when looking for the presence, location and type of hemorrhage on head computed tomography (CT), even for highly trained specialists. Therefore, it's a key challenge to achieve automated detection of intracranial hemorrhage on head CT with a promising detection accuracy at the examination level, leaving an urgent and critical issue to be addressed. In this paper, to accomplish this crucial task, we utilize a series of deep convolutional neural networks based on SE-ResNeXt50 and EfficientNet-B3 to simultaneously extract features from head CT and learn the classification for 5 subtypes of intracranial hemorrhage. The image dataset of head CT is provided by Radiological Society of North America through a featured competition on the Kaggle platform. The results show that the ensemble of SE-ResNeXt50 and EfficientNet-B3 trained with weighted multi-label logarithmic loss achieves an expert-level performance regarding the classification accuracy of intracranial hemorrhage. Our deep learning model obtain a score 0.0548 on the test set, which locates at the top 4% among all teams and can get a silver medal prize in the Kaggle competition.

CCS Concepts

• **Applied computing→Engineering**

Keywords

Intracranial hemorrhage; head computed tomography; automated detection; deep learning; computer aided diagnosis.

1. INTRODUCTION

Intracranial hemorrhage refers to bleeding that occurs inside the cranium. Such disease is a serious health problem requiring rapid and often intensive medical treatment. For example, intracranial hemorrhages account for approximately 10% of strokes (fifth-leading cause of death) in the U.S. [1]. To medically deal with the intracranial hemorrhage, identifying the location and type of any hemorrhage present is a critical step in treating the patient. For that purpose, head computed tomography (CT), the workhorse medical imaging modality used worldwide [2], is often utilized to help diagnose the intracranial hemorrhage. However, due to the limited quality of these gray-scale CT images, the diagnosis process of intracranial hemorrhage is complicated and often time consuming when looking for the presence, location and type of hemorrhage, even for highly trained specialists.

On the other hand, the past few years have been witness to an unprecedented revolution in computer vision techniques, especially the rapid development of deep learning methods [3]-[7] driven by big data. Extensive research works have demonstrated that convolutional neural network (CNN) has an incredible ability to extract clinically important information from medical images [8]-[10]. As a consequence, deep learning techniques have recently possessed a wide array of applications in medical treatments, such as classification of skin cancer [8], diagnosis of diabetic retinopathy [9], and detection of metastases [10]. That is, there is a tremendous potential using deep learning to achieve automated detection of intracranial hemorrhage on head CT with a promising detection accuracy at the examination level.

Recently, in collaboration with members of the American Society of Neuroradiology and MD.ai, a large-scale and high-quality image dataset of head CT images of intracranial hemorrhage has been provided by the Radiological Society of North America (RSNA) [11]. On the basis of this high-quality dataset, RSNA has also hosted a featured machine learning competition on the Kaggle platform [11], which aims to build an algorithm to detect acute intracranial hemorrhage and its subtypes, in order to help the medical community quickly and effectively treat affected patients. In this paper, we explore the applications of modern machine learning techniques based on the up-to-date CNN architectures for that purpose. By combing the results of deep neural networks based on the SE-ResNeXt50 [5], [6] and EfficientNet-B3 [7], we achieve the automated detection of intracranial hemorrhage on head CT images with an excellent performance regarding the classification accuracy. In particular, our deep learning model obtains a score 0.0548 on the test set, which locates at the top 4% among all teams and can get a silver medal prize in the Kaggle competition [11].

The rest of the paper is organized as follows. In Section 2, the image dataset provided by RSNA will be described in detail. In Section 3, we briefly introduce the CNN models based on SE-ResNeXt50 and EfficientNet-B3 backbones. The pipeline of our algorithm is demonstrated in Section 4, where the training schedule is also given. The classification performance of our model is described in Section 5. Finally, we make a conclusion based on our results in Section 6.

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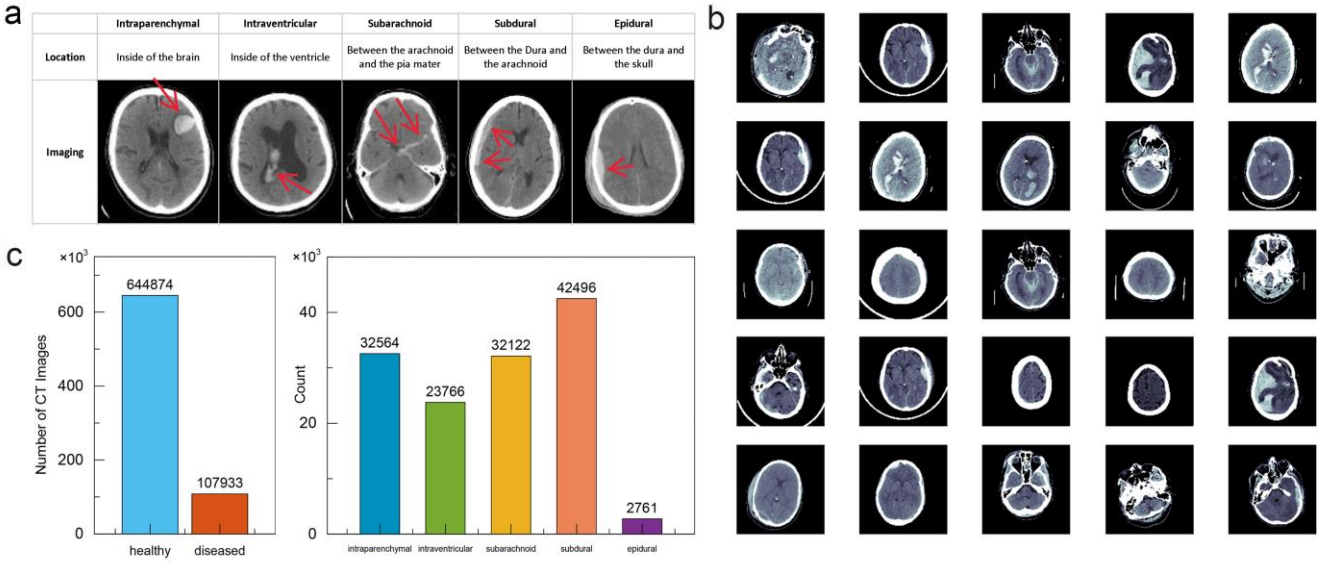


Figure 1. (a) Schematic diagrams of five intracranial hemorrhage subtypes, where their locations are highlighted by the red arrows. (b) Representative head CT images of five subtypes in the RSNA dataset. From top to down: intraparenchymal, intraventricular, subarachnoid, subdural, and epidural. (c) Data distribution of image labels in the training set. Left: label numbers for healthy and diseased patients. Right: label numbers of five subtypes.

2. DATA DESCRIPTION

2.1 Overview of RSNA Dataset

The large-scale and high-quality image dataset of head CT images of intracranial hemorrhage provided by RSNA contains 752807 images in the training set and 121232 images in the test set. All images are stored in a special format known as DICOM files, which contain a combination of header metadata as well as underlying raw image arrays for pixel data (512 ×512). While all acute hemorrhages appear dense (i.e. white) on CT images, the information in DICOM files will help Radiologists determine the subtype of hemorrhage.

Intracranial hemorrhage contains five subtypes (See Figure 1a), i.e., *intraparenchymal*, *intraventricular*, *subarachnoid*, *subdural*, and *epidural*. A brief introduction of such five subtypes is summarized here [1]. As demonstrated by the representative head CT images in Figure 1b, intraparenchymal hemorrhage is blood that is located completely within the brain itself; intraventricular or subarachnoid hemorrhage is blood that has leaked into the spaces of the brain that normally contain cerebrospinal fluid (the ventricles or subarachnoid cisterns). Extra-axial hemorrhages are blood that collects in the tissue coverings that surround the brain (e.g. subdural or epidural subtypes). Patients may exhibit more than one type of cerebral hemorrhage, which may appear on the same image.

2.2 Statistical Distribution of Image Labels

The labels of images in the training set are provided as multiple labels, one for each of five sub-types of hemorrhage, plus an additional label for *any*, which should always be true if any of the sub-type labels is true. As a consequence, the detection of intracranial hemorrhage on head CT is a multi-label classification challenge.

As demonstrated in the left of Figure 1c, RSNA dataset has 644874 images for patients with no hemorrhage (i.e. healthy) and 107933 images for patients with at least one subtype of hemorrhage (i.e. diseased) in the training set. This means that there exist much more zero occurrences than positive target values in the dataset. On the other hand, in the right of Figure 1c we show the distribution of labels of each subtype, which suggests a relatively high class imbalance, especially for the *epidural* (2.6% in diseased patients).

3. MODELS

In this paper, we utilize the SE-ResNeXt50 and EfficientNet-B3 as CNN backbones and weighted multi-label logarithmic loss as the loss function to train our deep learning model. Therefore, in this section, we will introduce these state-of-the-art CNN architectures and the loss function in detail.

3.1 SE-ResNeXt

SE-ResNeXt is the combination of ResNeXt network [5] and squeeze-and-excitation (SE) block [6]. ResNeXt is a ResNet-based architecture. ResNet was one of the most groundbreaking works in the computer vision community in recent years [4]. The core innovation of ResNet is introducing a so-called "identity shortcut connection" that skips one or more layers, as shown in the left of Figure 2a. By reusing activations from a previous layer until the adjacent layer learns its weights, the problem of vanishing gradients can be effectively avoided in ResNet. Such design makes it possible to train up to hundreds or even thousands of layers and still achieves compelling performance. As an improvement of ResNet, ResNeXt adapts aggregated transformations to the second convolution layer of each bottleneck block (See right of Figure 2a). The aggregated transformation can be mathematically presented as $\mathcal{F}(\mathbf{x}) = \Sigma_{i=1}^{C} \mathcal{T}_i(\mathbf{x})$ where $\mathcal{T}_i(\mathbf{x})$ can be an arbitrary function. C is the size of the set of transformations to be aggregated, and usually referred as cardinality.

In addition, a squeeze-and-excitation (SE) block (See Figure 2b) is applied at the end of each non-identity branch of residual block

of ResNeXt, as demonstrated in Figure 3c. The squeeze operator is for global information embedding, which utilizes the global average pooling to generate channel-wise statistics. Extensive works and experiments have proved that significant performance improvement can be induced by the introduction of SE blocks into ResNeXt architectures. SE-ResNeXt50 has a top-5 error of 5.49% which is superior to both its direct counterpart ResNeXt50 (5.90% top-5 error) as well as the deeper ResNeXt101 (5.57% top-5 error), a model which has almost twice the total number of parameters and computational overhead [6].

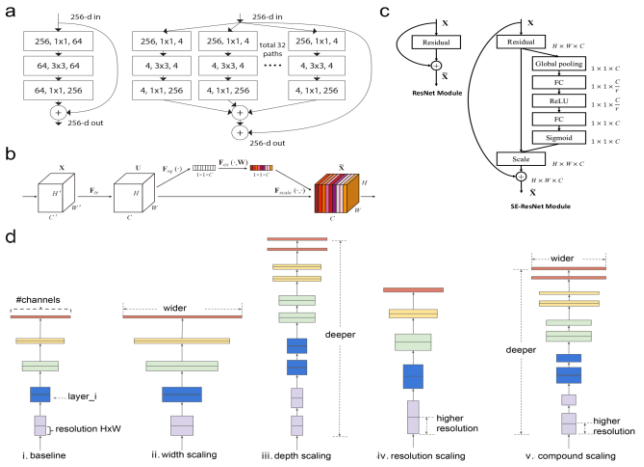


Figure 2. (a) The block of ResNet (left) [4] and the block of ResNeXt (right) with cardinality = 32 [5]. (b) The squeeze-and-excitation block. (c) Schematic diagram of the SE-ResNet module, where the SE block transformation is taken to be the non-identity branch of a residual module [6]. (d) Schematic diagram of model scaling used in EfficientNet. (i) baseline network example; (ii)-(iv) are conventional scaling that only increases one dimension of network width, depth, or resolution; (v) compound scaling method that uniformly scales all three dimensions with a fixed ratio [7].

3.2 EfficientNet

EfficientNet is a novel CNN model achieving both higher accuracy and better efficiency over existing CNNs, reducing parameter size and FLOPS by an order of magnitude [7]. In particular, in the high-accuracy regime, EfficientNet-B7 reaches state-of-the-art 84.4% top-1 / 97.1% top-5 accuracy on the ImageNet dataset, while being 8.4x smaller and 6.1x faster on CPU inference than the previous G-pipe [12]. The architecture of EfficientNet originates from a novel model scaling method, which is recently proposed by Google Research. As shown in Figure 2d, unlike conventional approaches that arbitrarily scale network dimensions, such method utilizes a simple yet highly effective compound coefficient to scale up CNNs in a more structured manner. The dimension of width, depth, and resolution are uniformly scaled with a fixed set of scaling coefficients w, d, and r respectively. The scaling coefficients can be expressed as $d = \alpha^{\phi}$, $w = \beta^{\phi}$, $r = \gamma^{\phi}$, where ϕ is the compound coefficient determining the network size, and $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, $\alpha \geq 1$, $\beta \geq 1$, $\gamma \geq 1$. The structure of baseline network EfficientNet-B0 (i.e., $\phi = 0$) is given in Table I, where $\alpha = 1.2$, $\beta = 1.1$, and $\gamma = 1.15$ (obtained by a small grid search).

Table 1. Summary of EfficientNet-B0 baseline network [7]

Stage	Operator	Resolution	Channels	Layers
1	Conv3x3	224×224	32	1
2	MBConv1, k3x3	112×112	16	1
3	MBConv6, k3x3	112×112	24	2
4	MBConv6, k5x5	56×56	40	2
5	MBConv6, k3x3	28×28	80	3
6	MBConv6, k5x5	14×14	112	3
7	MBConv6, k5x5	14×14	192	4
8	MBConv6, k3x3	7×7	320	1
9	Conv1x1 & Pooling & FC	7×7	1280	1

3.3 Loss Function

In this work, we use the weighted multi-label logarithmic loss to train our model and score the model performance. Briefly, for each subtype s for an image sample n, the standard logarithmic loss $l_{n,s}$ is first taken as follow:

$$l_{n,s} = t_{n,s} \cdot \ln(p_{n,s}) + (1 - t_{n,s}) \cdot \ln(1 - p_{n,s}), \quad (1)$$

where $p_{n,s}$ is the predicted probability that sample n belongs to subtype s, $t_{n,s}$ is the label of sample n (0 or 1). The total loss l_n for sample n can be calculated by:

$$l_n = \sum_s w_s \cdot l_{n,s}, \quad (2)$$

where w_s represents the weight for the subtype s. In order to make model pay more attention on identifying the presence of intracranial hemorrhage, we set $w_s = 2$ for label *any* and $w_s = 1$ for the 5 subtypes of hemorrhage. The final loss function L can thereby be written as:

$$L = -\sum_n l_n / N, \quad (3)$$

where N is total number of images in the training set.

4. PIPELINE AND TRAINING SCHEDULE

The schematic diagram of the deep learning pipeline in our works is shown in Figure 3. Randomly cropped 384×384 patches from the original images are used as the input with left and right flips as the data augmentation. After passing through the CNN backbone (SE-ResNeXt50 or EfficientNet-B3), the input image is decoded as a numerical vector, which is subsequently classified by the following full-connected layers. The last layer has 6 nodes with sigmoid activation functions, representing 6 classes in this classification task respectively. The final prediction p_n of image sample n is a linear combination of the results of SE-ResNeXt50 $p_{n,s}$ and EfficientNet-B3 $p_{n,e}$, written as:

$$p_n = \epsilon p_{n,s} + (1 - \epsilon) p_{n,e}, \quad (4)$$

where ϵ is determined through the grid search.

The training schedule is briefly summarized here. The CNN backbone is initialized with the ImageNet weights, while the full-connected layers are initialized using the He initialization [13]. Loss function is the weighted multi-label logarithmic loss

described in section 3.3. The model is trained by the back-propagation algorithm with batch gradient descent (batch size 22) and adam optimizer with the weight decay 0.01 [14]. Learning rate (LR) at the very beginning is set as 6×10^{-5}. During the training process, we reduce the LR with a factor 2/3 (LR = LR × factor) at the ending of each epoch. The training process lasts for 4 epochs. The time cost of training one epoch is about 16 hours on a single RTX 2080Ti.

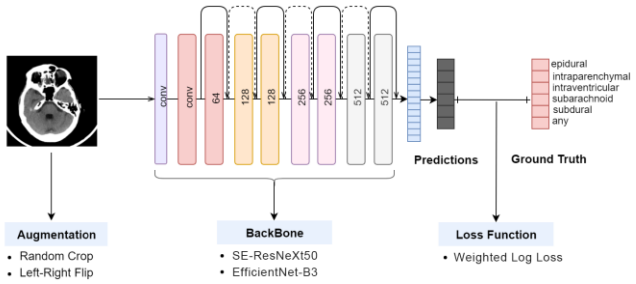


Figure 3. Schematic diagram of the deep learning pipeline. The augmentation is random crop and left-right flip. The CNN backbones are SE-ResNeXt50 and EfficientNet-B3. The loss function is the weighted multi-label logarithmic loss.

In addition, K-Fold cross validation [15] and pseudo-label technique [16] are used to improve the model performance. K-Fold cross validation splits the dataset into a *K* number (*K*=10 in this work) of folds, where each fold is used as the validation set while all other folds are used as the training set. The cross validation is repeated *K* times. The K-averaged results are used to get a better estimate of model prediction performance. On the other hand, pseudo label is a wide-used semi-supervised learning approach to improve model performance when the amount of unlabeled data is considerable. When applying such techniques, pseudo labels, which are the class with the maximum predicted probability, are used as if they were true labels. In our works, we set the pseudo label as the true label if $p_n > 0.995$ or $p_n < 0.005$, which adds 52260 images into the training set.

5. RESULTS

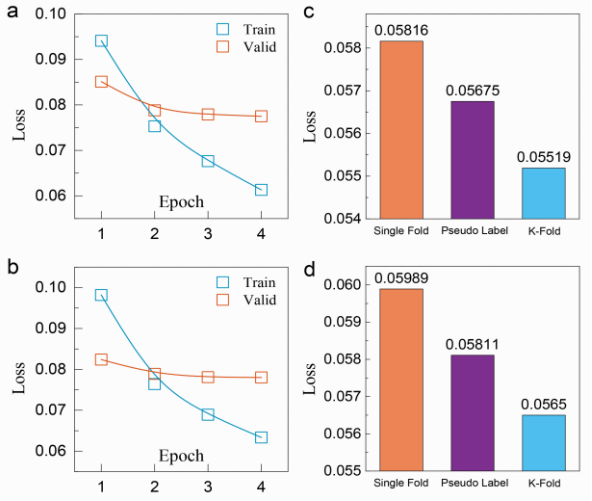


Figure 4. (a)-(b) Evolutions of train loss, valid loss of SE-ResNeXt50 (a) and EfficientNet-B3 (b) during the training process respectively. (c)-(d) Scores of SE-ResNeXt50 (c) and EfficientNet-B3 (d) when training with a single fold, pseudo label, and K-Fold cross validation respectively.

To visualize the results, the epoch dependences of train and valid loss of SE-ResNeXt50 and EfficientNet-B3 during the training process are given in Figures 4a and 4b, respectively. Best model usually emerges at the 4th epoch as indicated by the valid loss. The corresponding scores (weighted multi-label logarithmic loss) of best models on the test set (calculated by the Kaggle platform) are given in Figures 4c and 4d, respectively. For the model based on SE-ResNeXt50, the score for a single fold is near 0.0582, which can be improved to 0.0568 by utilizing the pseudo-label technique. By applying a K-Fold cross validation, the K-averaged result can reach a score 0.0552 (Figure 4c). Moreover, for the model with an EfficientNet-B3 backbone, the score of the K-averaged result is 0.0565 (Figure 4d). The final predictions are determined by Eq. (4) with $\epsilon = 0.6$, leading to a score 0.0548 on the test set. Such score locates at the top 4% among all teams and can get a silver medal prize in the Kaggle competition. That is, the ensemble model has achieved the accuracy performance very close to that of manual annotations given by highly trained specialists.

6. CONCLUSION

In summary, by utilizing a series of deep convolutional neural networks, we successfully achieve the goals to automatically detect the presence and classify the type of intracranial hemorrhage on head CT. The results show that the ensemble of CNN models based on SE-ResNeXt50 and EfficientNet-B3 trained with weighted multi-label logarithmic loss achieves an expert-level performance regarding the classification accuracy of intracranial hemorrhage. Our model obtain a score 0.0548 on the test set, which locates at the top 4% among all teams and can get a silver medal prize in the Kaggle competition. The deep learning model proposed in this paper can help the medical community quickly and effectively treat affected patients. However, such model cannot yet identify the location of intracranial hemorrhage on head CT. Therefore, the further step in the future is to explore the applications of object detection [17] and semantic segmentation [18] techniques in this field, which can potentially improve the efficiency of hemorrhage treatment significantly and save more time for patients requiring immediate surgery.

7. ACKNOWLEDGMENTS

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